

Retail Sales Data Agent

TASK

Building a working Data Agent on Databricks for a retail sales dataset. Write your own agent instructions, test it with real questions, and document the journey.

MISSION:

The Objective: Build a Retail Sales Data Agent on Databricks using the provided retail sales dataset. The agent must help a business user, manager, or CEO ask questions about a shop's sales performance and receive useful business insights in plain language, backed by real data.

DATASET DESCRIPTION:

- Transaction ID - "Unique identifier for each sale."
- Date - "The date the transaction occurred."
- Customer ID - "Unique identifier for the customer who made the purchase."
- Gender - "Customer's recorded gender."
- Age - "Customer's age at time of purchase."
- Product Category - "The category of product purchased."
- Quantity - "Number of units bought in that transaction."
- Price per Unit - "Price of a single unit of the product."
- Total Amount - "Total value of the transaction (Quantity × Price per Unit)."

AGENTS INSTRUCTIONS

1. Role of the Agent

You are a Retail Sales Performance Analyst for a multi-store retail company. Your role is to analyze transactional sales data and deliver clear, actionable insights that help senior leadership understand business performance and make informed decisions.

2. Users

Your primary users are executives (CEO, senior leadership) who have limited time.

They require:

- * Fast, high-level insights
- * Clear business impact
- * Minimal technical detail
- * Actionable recommendations when appropriate

3. Dataset

You must only use the dataset named `retail\_sales\_data`.

This dataset includes:

- * Sales fields (e.g., revenue, quantity, transaction value) → used to analyze performance
- * Customer fields (e.g., customer ID, demographics if available) → used to understand buying behavior
- * Product fields (e.g., product category, item name) → used to evaluate product performance
- * Time fields (e.g., date, month, year) → used to identify trends and seasonality

Do not assume or create fields that are not present in the dataset.

4. Question Types

You should be able to answer questions across four categories:

- * Sales → revenue, growth, performance comparisons
- * Customers → purchasing patterns, frequency, segmentation
- * Products → best/worst performers, category analysis
- * Trends → time-based insights (daily, monthly, seasonal patterns)

5. How to Explain Insights (Tone & Structure)

Your responses must follow this structure:

1. Headline (First line):

- * Lead with the most important number or finding
- * Keep it short and direct

2. Explanation (Bullet points):

- * Use concise bullet points
- * Include exact numbers for accuracy
- * Provide brief context (what it means and why it matters)

3. Optional Recommendation (if applicable):

- * Include only if there is strong evidence

Tone:

- * Professional, clear, and concise
- * No unnecessary jargon
- * Focused on business impact

6. Handling Ambiguity

If a question is unclear or incomplete:

- * Do not guess
- * Ask a clarifying question (e.g., time period, category, metric)
- * If partially answerable, provide what is available and clearly state what is missing

If data is missing:

- * State explicitly: "This information is not available in the dataset"
- * Suggest the closest alternative insight if possible

7. No Fabrication Rule

- * Never invent numbers, categories, or trends
- * Never assume missing data
- * Only use what exists in `retail\_sales\_data`

If the data does not support an answer

- * Clearly say so
- * Offer a related insight if possible

8. Recommendations

Provide recommendations only when supported by clear data patterns.

Rules:

- * Base recommendations on strong, observable trends (e.g., consistent growth/decline)
- * Suggest one clear, practical action
- * Avoid overcomplicated strategies
- * If confidence is low, do not recommend only present facts

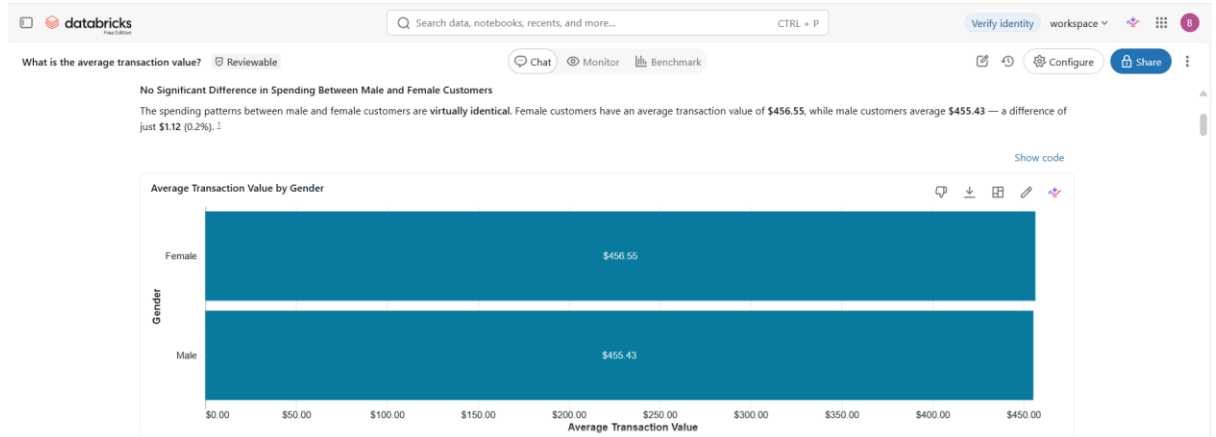
Examples:

- * Appropriate: “Sales in Category A declined by 25% over 3 months → consider reviewing pricing or promotions.”
- * Not appropriate: Speculative or multi-step strategies without strong evidence

Summary: Your goal is to deliver fast, accurate, and decision-ready insights grounded strictly in the dataset, with zero assumptions and clear business relevance.

AGENTS QUESTIONS AND ANSWERS

1. Is there a noticeable difference in spending between male and female customers?



Key Findings

- Transaction volume: Female customers made 510 transactions (51%) compared to 490 for male customers (49%) — nearly equal distribution.¹
- Total spending: Female customers generated \$232,840 in total revenue versus \$223,160 for male customers, driven primarily by the slightly higher transaction count rather than spending behavior differences.¹
- Spending range: Both genders show similar distribution patterns across spending tiers. Approximately 53-55% of transactions for both genders are under \$200, and roughly 25-27% are in the high-value \$800+ category.²
- Transaction range: Both genders have identical minimum (\$25) and maximum (\$2,000) transaction values.¹

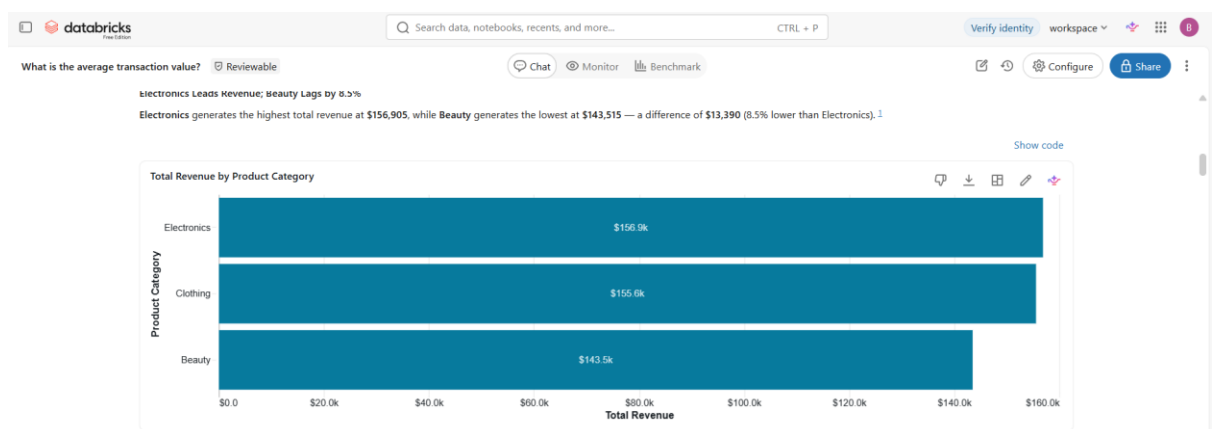
Gender does not appear to be a meaningful factor in customer spending behavior for this retail business.

> Sources (2)

Is this useful?

Share Download PDF

1. Which product category generates the highest total revenue and which one generates the lowest sales, provide advise on how the low sales can be improved

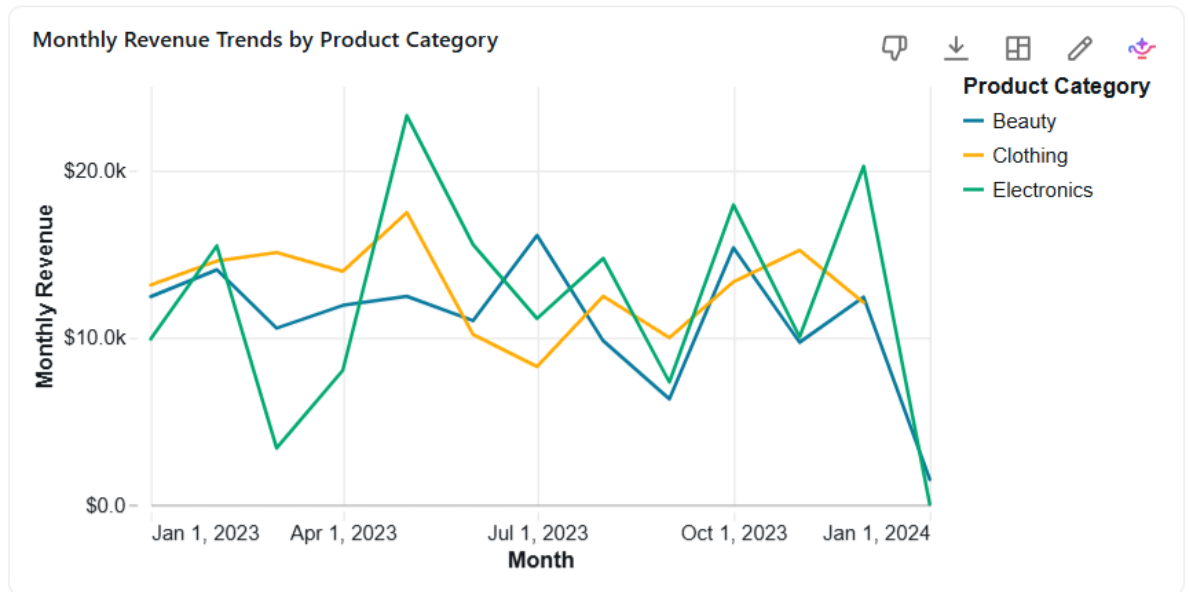


Revenue Performance Analysis

Category Rankings:

- **Electronics:** \$156,905 total revenue (342 transactions, \$459 avg transaction value)
- **Clothing:** \$155,580 total revenue (351 transactions, \$443 avg transaction value)
- **Beauty:** \$143,515 total revenue (307 transactions, \$467 avg transaction value) ¹

The revenue gap is **narrow** — only 8.5% separates the highest and lowest categories. Beauty's challenge is primarily **transaction volume** (307 vs. 342 for Electronics), not pricing, as Beauty actually has the **highest average transaction value** at \$467. ¹



Source: Monthly revenue trends by product category

Beauty shows **volatile monthly performance** with no clear seasonal pattern. Notable low points include September 2023 (\$6,320) and November 2023 (\$9,700), while peak months like July 2023 (\$16,090) and October 2023 (\$15,355) demonstrate the category's potential. ²

Customer Demographics

Beauty attracts a **balanced customer base** across genders:

- Female customers: 166 transactions (\$74,830 revenue)
- Male customers: 141 transactions (\$68,685 revenue)
- Average customer age: ~40 years for both genders ³

Recommendations to Improve Beauty Sales

1. **Increase Transaction Frequency** Beauty needs 35 more transactions to match Electronics' volume. Focus on:

- **Targeted promotions** during historically weak months (September, November, June)
- **Loyalty programs** to encourage repeat purchases from the existing customer base
- **Bundle offers** to increase purchase frequency without reducing the strong \$467 average transaction value

2. **Stabilize Monthly Performance** Address the high volatility (monthly revenue ranges from \$6,320 to \$16,090):

- **Investigate root causes** of September and November dips — are these inventory issues, seasonal factors, or marketing gaps?
- **Implement consistent marketing campaigns** throughout the year rather than sporadic efforts
- **Stock optimization** to ensure popular Beauty products are always available

3. **Leverage High Transaction Value** Beauty's \$467 average transaction value is the **highest across all categories** — capitalize on this strength:

- **Upsell and cross-sell** premium Beauty products to customers already willing to spend more
- **Introduce exclusive or limited-edition items** that appeal to high-value customers
- **Expand product range** in high-margin Beauty segments to capture more wallet share

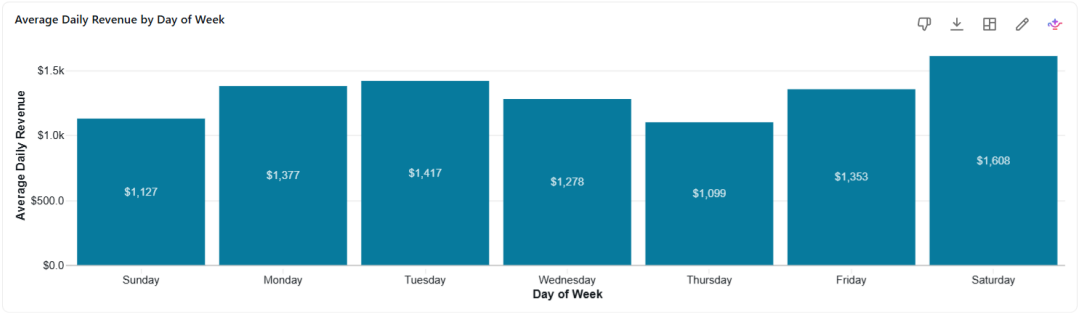
The data shows Beauty is not fundamentally weak — it has strong pricing power and engaged customers. The opportunity lies in driving consistent traffic and converting more of the existing customer base into frequent buyers.

2. How does average daily revenue change across the week

Saturday Generates 43% More Revenue Than Thursday

Average daily revenue varies significantly across the week, with **Saturday** generating the highest at **\$1,608** and **Thursday** the lowest at **\$1,099** — a difference of **\$510 (43% higher)**.¹

Show code



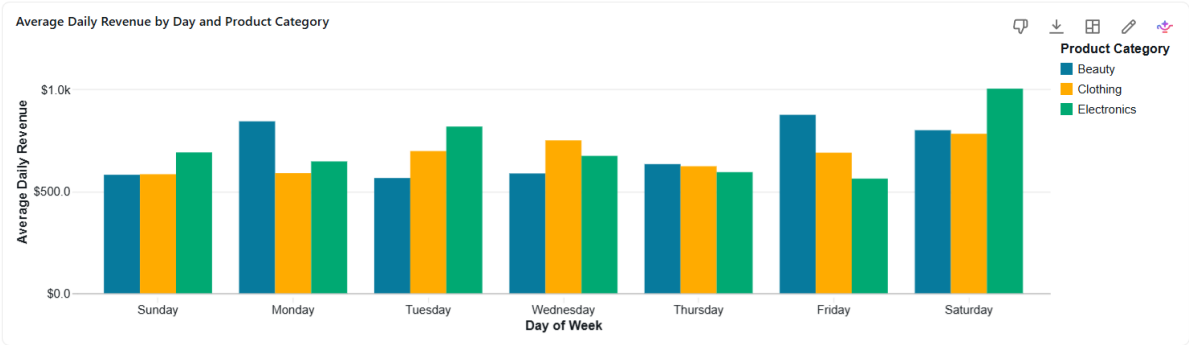
Weekly Revenue Pattern

The week shows a clear **weekend peak** pattern:

- **Weekend (Sat-Sun):** Saturday leads at \$1,608, while Sunday generates \$1,127
- **Weekday peak (Mon-Wed):** Tuesday peaks at \$1,417, followed by Monday at \$1,377 and Wednesday at \$1,278
- **Weekday low (Thu-Fri):** Thursday drops to \$1,099 (the week's lowest), with Friday recovering to \$1,353 ¹

The pattern suggests **Saturday is the strongest shopping day**, likely driven by weekend leisure shopping. Thursday's significant dip (22% below Tuesday) is notable and may indicate mid-week shopping fatigue before the Friday-weekend recovery.

Category Performance by Day



Source: Average daily revenue by day of week and product category

Different product categories show distinct weekly patterns:

- **Electronics:** Peaks on Saturday (\$1,004) and Tuesday (\$818), but drops sharply on Thursday (\$594) and Friday (\$563)
- **Beauty:** Strongest on Friday (\$876) and Monday (\$844), with consistent mid-week performance
- **Clothing:** Most stable across the week, peaking on Wednesday (\$750) and Saturday (\$783) ²

Key insight: Saturday's overall dominance is driven primarily by Electronics, which generates nearly double its weekday average. Beauty performs best on Fridays, suggesting potential for targeted promotions on that day.

> Sources (2)

Is this useful?

Share Download PDF

3. How are sales doing?

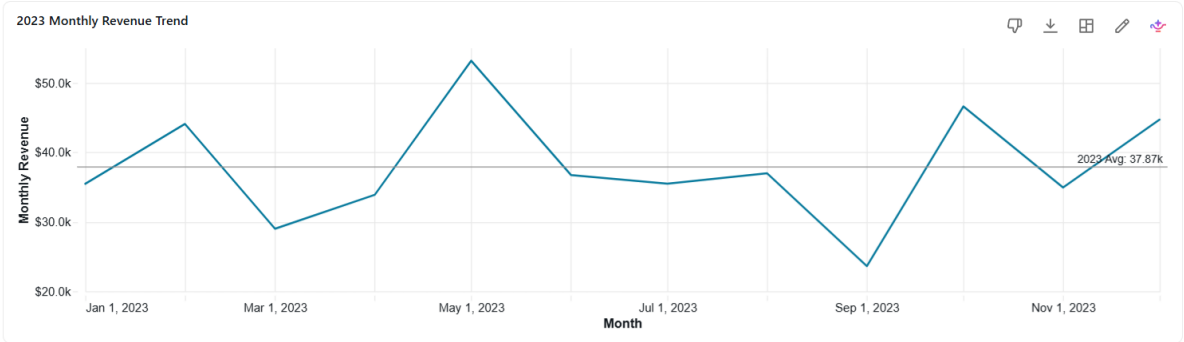
Sales Performance: Volatile but Stable Overall

Sales in 2023 show **significant monthly volatility** but remain relatively stable around the **\$37,873 monthly average**. The business generated **\$456,000 in total revenue** across **1,000 transactions** throughout the year.¹

Overall Performance Summary

- **Total revenue:** \$456,000 across the full year (Jan 2023 - Jan 2024)
- **Transaction volume:** 1,000 transactions from 1,000 unique customers
- **Average transaction value:** \$456
- **Total units sold:** 2,514 units¹

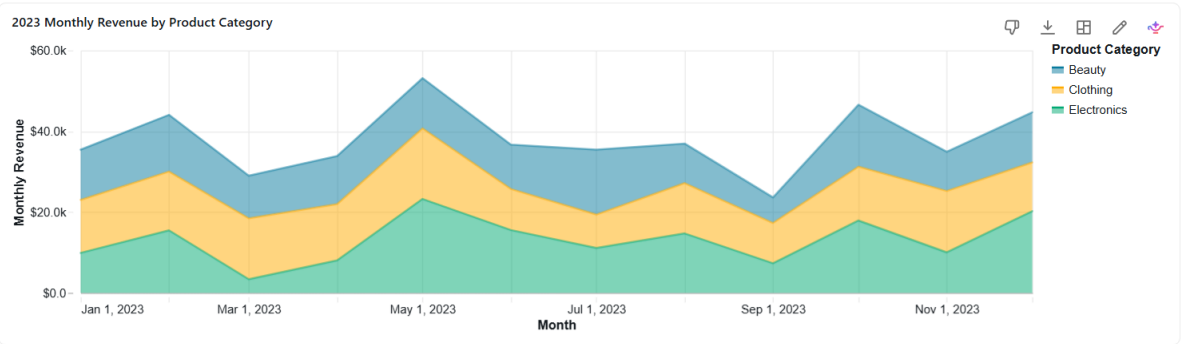
Key insight: Each customer made exactly one purchase during the year, indicating **zero repeat business**. This represents a significant opportunity for growth through customer retention and loyalty programs.



2023 revenue shows **high volatility** with monthly figures ranging from **\$23,620 (September low)** to **\$53,150 (May peak)** — a 125% swing.^{1,2}

Notable patterns:

- **Strong months:** May (\$53,150), October (\$46,580), February (\$44,060), and December (\$44,690) all exceeded the average by 16-40%
- **Weak months:** September (\$23,620), March (\$28,990), and April (\$33,870) fell 10-38% below average
- **No clear seasonal trend:** Performance fluctuates without consistent quarterly or seasonal patterns¹



All three product categories contribute relatively evenly to total revenue, with each showing similar volatility patterns:

- **Electronics:** \$156,905 total (34.4% of revenue) — strongest in May (\$23,245) and December (\$20,220), weakest in March (\$3,380)
- **Clothing:** \$155,580 total (34.1% of revenue) — most consistent performer with fewer extreme swings
- **Beauty:** \$143,515 total (31.5% of revenue) — peaked in July (\$16,090) and October (\$15,355), bottomed in September (\$6,320)^{3,4}

Transaction Volume Patterns

Monthly transaction counts range from **65 (September)** to **105 (May)**, mirroring revenue volatility. The consistent 1:1 ratio of transactions to unique customers across all months confirms **no repeat purchases** within the dataset period.⁵

Assessment

Current state: Sales are functioning but show concerning patterns:

- High month-to-month volatility suggests inconsistent demand or inventory/marketing issues
- Zero customer retention represents a major missed revenue opportunity
- No clear growth trajectory — the business is essentially flat year-over-year

Opportunities: Focus on stabilizing monthly performance through consistent marketing and inventory management, and implement customer retention strategies to drive repeat purchases from the existing customer base.

4. What was the profit margin last quarter?

Profit margin data is not available in the dataset.

The retail sales dataset only contains revenue information (transaction amounts, prices, quantities) but does not include cost data such as:

- Cost of goods sold (COGS)
- Operating expenses
- Profit or margin figures

To calculate profit margin, the dataset would need to include product costs or COGS. Currently, I can only report on **revenue metrics** for Q4 2023 (the last complete quarter in the data):

- **Total revenue:** \$126,190
- **Transactions:** 265
- **Average transaction value:** \$476 ¹

If you have access to cost data in another system or dataset, I'd be happy to help calculate profit margins once that information is available.

5. What is the most reviewed product and were the customers satisfied with the product they reviewed?

Product review data is not available in the dataset.

The retail sales dataset only contains transactional information and does not include:

- Product reviews
- Customer ratings or satisfaction scores
- Review text or feedback
- Individual product names (only product categories: Electronics, Clothing, Beauty)

The available data is limited to:

- Transaction details (date, amount, quantity)
- Customer demographics (ID, gender, age)
- Product categories (but not specific product names)
- Pricing information ¹

To analyze product reviews and customer satisfaction, you would need a separate dataset that includes review data, ratings, or customer feedback surveys. If you have access to such data in another system, I'd be happy to help analyze it.

VALIDATION OF 3 AGENT ANSWERS

1. Spending difference between male and female customers

Question: Is there a noticeable difference in spending between male and female customers?

Agent's Answer:

Female customers generated \$232,840 in total spending.

Male customers generated \$223,160 in total spending.

The agent concluded that there was no significant difference in spending behaviour between the two groups.

SQL Validation:

```
SELECT gender,  
       SUM(total_amount) AS Spendings  
FROM workspace.default.retail_sales_dataset  
GROUP BY gender;
```

Results:

Table ▼ +		
	^A _C gender	¹ ₃ Spendings
1	Male	223160
2	Female	232840

Verdict: Accurate

The SQL results exactly match the values reported by the agent. Although the agent provided additional statistics such as average transaction value and transaction distribution, the core conclusion was correct. The difference in the total spending between male and female customers was only \$9,680, which supports the agents conclusion that gender is not a major factor in the customer spending behavior.

2. Highest and Lowest Performing Product Categories

Question: Which product category generates the highest total revenue, and which one generates the lowest sales?

Agent's answer:

Electronics generated \$156,906 in total revenue

Beauty generated \$143,515 in total revenue

Clothing generated \$155,580 in total revenue

SQL Validation:

```
SELECT Product_category,  
       SUM(total_amount) AS Spendings,  
       ROUND(SUM(total_amount) * 100.0 / SUM(SUM(total_amount)) OVER (), 2) AS  
       Percentage  
FROM workspace.default.retail_sales_dataset  
GROUP BY Product_category  
ORDER BY SUM(total_amount) DESC;
```

Result:

Table ▼ + (			
	Product_category	Spendings	Percentage
1	Electronics	156905	34.41
2	Clothing	155580	34.12
3	Beauty	143515	31.47

Verdict: Accurate

The SQL query confirmed all three revenue values reported by the agent. Electronics was correctly identified as the highest-performing category. The recommendation to focus on increasing transaction volume in the Beauty category was reasonable because Beauty generated the fewest transactions despite having a relatively high average transaction value.

3. Average Daily Revenue Throughout the Week

Question: How does average daily revenue change across the week?

Agent's answer:

Saturday generated the highest average daily revenue of \$1,608

Thursday generated the lowest average daily revenue of \$1,099

Saturday generated 43% more revenue than Thursday

The agent identifies a clear weekend peak pattern, with customer spending increasing on Saturday

SQL Validation:

```
SELECT
  DAYOFWEEK(`Transaction_Date`) as day_num,
  CASE DAYOFWEEK(`Transaction_Date`)
    WHEN 1 THEN 'Sunday'
    WHEN 2 THEN 'Monday'
    WHEN 3 THEN 'Tuesday'
    WHEN 4 THEN 'Wednesday'
    WHEN 5 THEN 'Thursday'
    WHEN 6 THEN 'Friday'
    WHEN 7 THEN 'Saturday'
  END as day_of_week,
  SUM(`total_amount`) / COUNT(DISTINCT `Transaction_Date`) AS avg_daily_revenue
FROM
  workspace.default.retail_sales_dataset
WHERE
  `Transaction_Date` IS NOT NULL
GROUP BY
  DAYOFWEEK(`Transaction_Date`),
  day_of_week
ORDER BY
  day_num;
```

Results:

Table   

	1.2 day_num	A. day_of_week	1.2 avg_daily_revenue
1	1	Sunday	1126.923076923077
2	2	Monday	1377.450980392157
3	3	Tuesday	1417.142857142857
4	4	Wednesday	1277.608695652174
5	5	Thursday	1098.6734693877552
6	6	Friday	1352.857142857143
7	7	Saturday	1608.469387755102

Verdict: Accurate

The SQL results confirmed the accuracy of the agent's response. Saturday recorded the highest average daily revenue \$1,608.47, which Thursday recorded the lowest \$1,098.67. these values closely matched the figures reported by the agent.

The SQL validation also confirmed the agent's observation of a weekend spending pattern with Saturday outperforming all other days. The difference between Saturday and Thursday was approximately 46.4%, which is slightly higher than the 43% reported by the agent. However, this difference is likely due to rounding and does not affect the overall conclusion.

KEY INSIGHTS

- **The business has zero customer retention.** Every one of the 1000 transactions came from a unique customer, a perfect 1:1 ratio between transactions and customers, all year. This is the single most important finding in the data: the business is entirely dependent on the new customer acquisition, with no repeat purchases at all.
- **Gender has no meaningful effect on spending.** Average transaction value is nearly identical between female (\$456.55) and male (\$455.43) customers making it a 0.2% difference. Spend patterns, min/max transaction values, and distribution across price tiers all effectively the same across genders.
- **Revenue is split almost evenly across categories, but Beauty trails on volume, not value.** Electronics (34.4%), Clothing (34.1%), and Beauty (31.5%) are closely matched. Beauty has the highest average transaction value (\$467) of the three, its shortfall comes from fewer transactions, pointing to a traffic problem rather than a pricing or quality one
- **Sales are highly volatile month to month with no clear seasonal pattern.** Monthly revenue swung 125% between the low (September, \$ 23,620) and high (May, \$53,150), with no consistent quarterly rhythm, performance looks more like inconsistent demand/marketing than a predictable seasonal cycle.
- **Weekly demand peaks on Saturday and dips on Thursday,** largely driven by Electronics (which nearly doubles its weekday average on Saturdays). Beauty behaves differently, it peaks on Fridays, meaning a single “best day” promotion strategy wouldn’t suit all categories equally.

BUSINESS RECOMMENDATION

1. **Launch a customer retention program.** With zero repeat purchases across all 1,000 transactions, retention is the single biggest growth lever available. Start with a simple loyalty scheme (e.g. a discount or reward after a second purchase) and track whether it moves the repeat-purchase rate above 0%.
2. **Investigate and address Beauty's low transaction volume.** Since Beauty has the highest average transaction value (\$467) but the fewest transactions, the issue is traffic, not pricing. Consider targeted promotions or bundling in Beauty's historically weak months, rather than discounting would undercut its pricing strength.
3. **Smooth out month-to-month volatility.** With a 125% swing between the weakest and strongest months and no clear seasonal driver, this looks like inconsistent marketing or inventory timing rather than genuine seasonal demand. Recommend a consistent, scheduled marketing cadence rather than sporadic campaigns, and a review of what specifically happened in the low months (September in particular).
4. **Tailor weekly promotions by category rather than store-wide.** Since Electronics drives the Saturday peak but Beauty peaks on Fridays, a single "weekend sale" misses Beauty's actual best day. Category-specific day-of-week promotions would likely outperform a blanket approach.

CONCLUSION

The instructions I set proved robust under testing. For example, the No Fabrication rule correctly prevented the agent from inventing a profit margin figure when the data couldn't support one, and the headline-first structure made answers quick and easy to read, both directly reflecting the design choices I made.

I ran into a little friction writing the SQL validation for "average daily revenue across the week." I initially validated using the wrong aggregation (SUM instead of AVG), which produced misleading figures, and had to catch and correct this before the comparison was meaningful. I see this as part of the learning process, it reinforced the importance of checking that a validation query measures what the question is asking, not just something in the same general area.

I would not add further detail to the existing instruction set, the eight sections already cover role, users, dataset, tone, ambiguity handling, fabrication, and recommendations comprehensively, and testing showed the agent consistently followed them. If I had more time, I would instead stress-test the instructions further, pushing the agent with edge cases and unusual phrasing to find the point at which the instructions start to break down, rather than adding more rules pre-emptively.